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ORIGINAL RESEARCH ARTICLE

Prediction of Maternal and Fetal Acyclovir, Emtricitabine, Lamivudine, and Metformin Concentrations during Pregnancy Using a Physiologically Based Pharmacokinetic Modeling Approach

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Abstract

Background Concerns over maternal and fetal drug exposure during pregnancy highlight the need for improved understanding of drug distribution to the fetus through the placental barrier.

Objective Our objective was to predict maternal and fetal drug disposition using a physiologically based pharmacokinetic (PBPK) modeling approach.

Methods We used the detailed maternal–placental–fetal PBPK model within the Simcyp Simulator V20 to predict the maternal and fetal drug exposure of acyclovir, emtricitabine, lamivudine, and metformin during pregnancy and at delivery. The dynamic model includes gestational changes to the maternal, fetal, and placental physiological parameters. Placental kinetics were parameterized using published ex vivo data for these four compounds. Amniotic data were included where available. PBPK predictions were compared with the observed data using twofold criteria.

Results Maternal–fetal PBPK models were developed completely from the bottom up without any parameter adjustments. The PBPK model-predicted exposures matched the observed maternal and umbilical exposure for acyclovir (six maternal studies, all of which all reported umbilical exposure), emtricitabine (six maternal studies, of which four reported umbilical exposure), lamivudine, (five maternal studies, of which four reported umbilical exposure), and metformin (seven studies, of which six reported umbilical exposure). Predicted pharmacokinetic parameters were within twofold of the observed values.

Conclusion Integration of fetal and maternal system parameters within PBPK models, together with experimental data from ex vivo placental perfusion studies, facilitated and extended the application of the pregnancy PBPK model. Such models can also be used inform clinical trials and maternal/fetal risk assessment following maternally administered drugs or unintended exposure to environmental toxicants.

1 Introduction

Various physiological and biochemical changes in many maternal organs during pregnancy can affect drug pharmacokinetics. For drugs that can cross the placental barrier, changes in maternal exposure may affect fetal exposure, depending on the degree of drug distribution into the fetal circulation. Although most drug labels carry warnings about or contraindications to their use during pregnancy, maternal and fetal drug exposure to these drugs remains extensive. Studies have shown that approximately 24% of drugs are prescribed off label to pregnant women in the UK (16% of which were considered high risk) [1], about 28% of drugs in France (22% of which were considered high risk) [2], and about 23% of drugs in the USA (94% of which were administered during the third trimester) [3]. Virtually all drugs administered to a pregnant woman reach the fetus at widely variable levels. However, unlicensed or off-label drug administration may be necessary if the clinical need cannot be met by licensed medicines. Such use should be supported by appropriate evidence that balances the benefits and risks associated with fetal drug exposure. In certain fetal disorders, such as fetal arrhythmias, lung maturation, and other conditions requiring drug delivery to the fetus, it is also important to weigh the benefits of fetal drug therapy against the risks of maternal side effects.

Sampling the placenta, fetal organs, or systemic circulation during pregnancy for pharmacokinetic evaluation can be challenging and unethical. Experimentally, fetal drug

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Key Points

Maternal drug disposition of renally eliminated compounds depends on changes in maternal renal function, which can affect maternal and fetal exposure.

Integration of maternal and fetal gestational-dependent parameters within physiologically based pharmacokinetic (PBPK) models facilitates the prediction of both maternal and fetal drug exposure. Incorporating the physiological parameters using continuous growth functions as time-dependent covariates allows predictions to be performed at any gestational weeks.

The investigated drugs were mainly eliminated through the kidneys. Predictions at different gestational weeks indicated the adequacy of the model and the incorporated changes in renal function during pregnancy.

The maternal–fetal PBPK models developed in this study can be used to support the use of a PBPK-based approach in maternal, placental, and fetal safety assessments.

exposure remains difficult to quantify, as the fetus and placenta are not readily accessible for sampling until delivery. Sampling of the placenta and the umbilical cord at birth provides information on placental and fetal exposure; however, such measurements are still challenging because only one sample can be obtained per subject and the sampling time is relative to the last maternal dose intake. Various animal models have been used to study placental drug transfer, but data are of poor translational value because of interspecies variability in placental structure and hemodynamics [4, 5]. On the other hand, an ex vivo human placenta perfusion experiment that uses a single placental cotyledon has been shown to be a useful model for studying the transplacental passage of various drugs [4].

Pregnancy physiologically based pharmacokinetic (PBPK) models that incorporate adequately detailed maternal and fetal physiological parameters are potential tools that can facilitate the assessment of fetal exposure when placental kinetics are also integrated in the model [4, 6, 7]. The applications of PBPK models to describe fetal exposure have been reported for many drugs, including emtricitabine, tenofovir, nevirapine, midazolam, theophylline, darunavir, dolutegravir, zidovudine, and acetaminophen (see Abduljalil and Badhan [8]). In most of these examples, transplacental transfer parameters were estimated from ex vivo human placenta perfusion experiments and then integrated within the pregnancy PBPK models using appropriate scaling from a single cotyledon to the whole placenta. Those models included at least one compartment to reflect the anatomy of the fetus. Model predictions agreed with the clinical observations, which support the use of a PBPK model as a potential tool to assess drug exposure in the fetus and placenta following maternal intake.

The aim of this study was to extend the previously published PBPK model within the Simcyp Simulator V20 to include the multicompartmental fetal and placental components. Four renally eliminated compounds—acyclovir, emtricitabine, lamivudine, and metformin—were used to assess the model's performance. These compounds were chosen because of the availability of observed data. PBPK predictions at different time points during gestation were compared with the clinically observed data.

2 Methods

2.1 General Settings

The pregnancy model within the Simcyp Simulator V20 was used for all predictions in the current study. The pregnancy model accounts for interindividual variability in physiological parameters during the whole gestational period. Increases and decreases in physiological parameters during pregnancy are incorporated in the model as continuous functions to allow predictions at different gestational ages. The model considers the continuous change of all physiological and biological parameters simultaneously over time, and within each subject, to account for any time-varying covariates.

The model includes the multicompartment fetal model coupled with the maternal model via a permeability-limited placental model. It includes previously reported physiological changes that occur during pregnancy, including maternal [9] and fetal physiology [10–13]. A perfusion-limited version of a PBPK model consisting of 14 compartments representing various fetal tissues was linked to the full Simcyp maternal PBPK model via the placenta, which in turn was represented by three compartments (Fig. 1). Amniotic fluid was represented by a single compartment. Nonlinear differential equations describing the structure of the fetus, the placenta, and the amniotic fluid compartment within the model have already been published [14]. These equations are based on the mass conservation law describing changes in drug concentration over time after maternal drug administration. However, the model is further expanded to include eight additional fetal tissues: spleen, pancreas, muscle, bone, adipose, heart, lung, and skin. Physiological parameters required for fetal PBPK model specifications and their changes during pregnancy were taken from a recent series of published meta-analyses of physiological, biological, and anatomical measurements considering the total body weight, height, surface area and gross body composition [10], organ

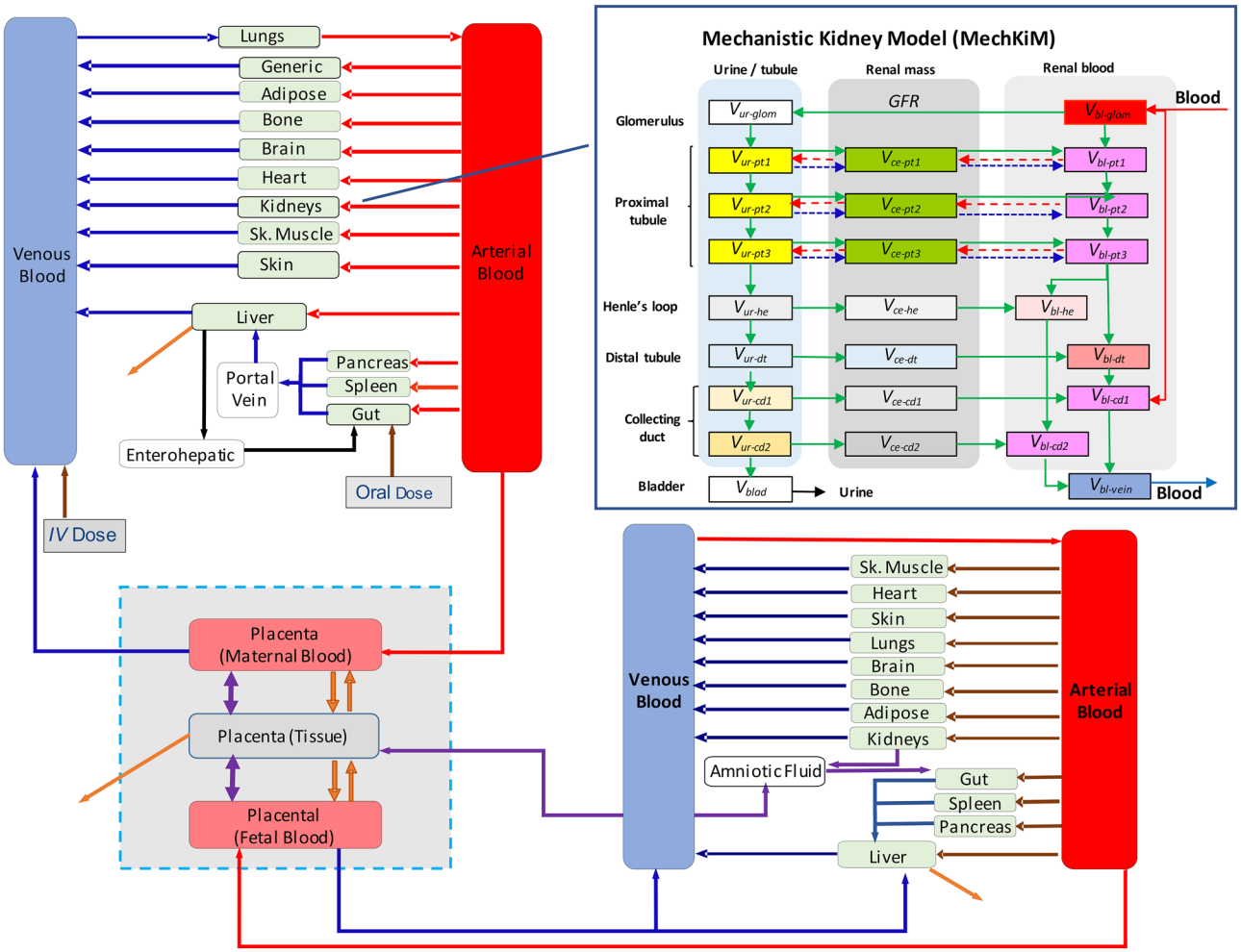


Fig. 1 Structure of the full maternal–placental–fetal model within the Simcyp Simulator. The single perfusion kidney model (default) was replaced with the mechanistic permeability-limited kidney (MechKiM) model to account for the transporter activity during pregnancy for acyclovir, lamivudine, and metformin. Physiological parameters in the MechKiM model, i.e., the renal blood flow, glomerular filtration rate, and, hence, intercompartmental flow as well as the OCT2 transporter activity are gestational dependent. *IV* intravenous, *Sk* skeletal, V_x volume of the x renal compartment, *bl* renal blood, *glom* glomerulus, *pt* proximal tubule (3 segments), *ce* renal cell, *hp* Henle's loop, *dt* distal tubules, *cd* collecting duct (2 segments), *blad* bladder

volumes and compositions [11], blood and binding components [13], and organ blood flows [12]. Equations describing longitudinal changes in fetal physiological parameter values were incorporated into the fetal PBPK model and coupled to the maternal PBPK model within the Simulator. The model predicts the gestational age-dependent unbound fractions of drugs in fetal and maternal circulation from the plasma concentration of binding protein in the fetal and maternal circulation at each gestational age (see Abduljalil et al. [15]).

The placental model includes published physiological data. The total volumes of the placental and maternal blood supply at different gestational weeks and their compositions have been reported previously [9], and the fetal blood supply through the umbilical vessels was reported in a more recent study [12]. The placenta is split into three compartments—maternal blood, placental cellular tissue, and fetal blood—to allow characterization of the drug transfer on both sides of the placental cell. The average maternal blood volume in the placenta is about 11.55% of the total placental weight, whereas the average fetal blood constituents are about 8.25% of the placental weight [16]. The remaining 80.2% represents the placental tissue. The tissue volume constitutes about 76.3% of the total placental weight [16], and this is further split into extracellular and intracellular volumes, representing about 56.3 and 11.1% of the placental weight, respectively. Since the placental volume depends upon gestational age, so does the absolute volume of these components. The current model does not differentiate between trophoblast and stroma. By default, the model predicts the unbound fraction of the drug in the placental

intracellular compartment according to the Rodgers and Rowland method [17]. These data were used to parameterize different component volumes of the placenta in the model. The permeability clearance input in the model is unit of mL/min/mL of placental volume. This approach is useful to allow scaling of the permeability to early gestational weeks based on the placental volume.

Amniotic fluid is mainly composed of water, and its volume increases with pregnancy progression, according to our previously published data [9]. The model assumes that the administered drug reaches the amniotic fluid via fetal renal clearance (CL_R) and re-enters the fetal circulation via fetal swallowing activity and intramembranous diffusion. These activities change the drug concentration in the amniotic fluid but do not cause any significant change in the volume of the amniotic fluid.

2.2 Model Building

For all evaluated compounds, distribution was defined using a full PBPK distribution model that accounted for different tissue volumes and flow rates. Drug tissue-to-plasma partition coefficients (Kps) were predicted within the Simulator for these compounds using the Rodgers and Rowland method [17]. Renal clearances of the drugs were described either using a passive process through glomerular filtration rate (GFR) or the mechanistic kidney model (MechKiM) to account for renal transporter kinetics [18], depending on data availability for each of the assessed compound. No fetal hepatic clearance was assumed for these compounds. Fetal CL_R was calculated with reference to an adult GFR of 121 mL/min [19] according to Eq. (1):

$$\text{fetal } CL_R \text{ (L/h/kg)} = \left(\frac{\text{Adult } CL_R \text{ (L/h)}}{\text{Fetal Bodyweight (Kg)}}\right) \cdot \left(\frac{\text{Urine Flow (mL/min)}}{\text{Adult GFR (mL/min)}}\right) \quad (1)$$

A fetal bodyweight of 3.5 kg at term was used to scale the clearance per kilogram.

At term, the fetus swallows an average of 400 mL of amniotic fluid per day [20]. Therefore, this value was used to describe the swallowing activity clearance in the model (i.e., $CL_{swallowing}$ [L/h/kg fetal weight] = 0.00476 L/h/kg fetal weight). Additionally, the intramembranous pathway transfers about 200–500 mL/day of fluid and solutes from the amniotic cavity to the fetal circulation across the amniotic membranes [21]. This flow has been calculated using 350 mL/day and normalized to kilogram of fetal body weight at term (i.e., $CL_{intramembranous}$ = 0.00417 L/h/kg). The combined fetal $CL_{swallowing}$ and $CL_{intramembranous}$ was incorporated into the model ($CL_{swallowing}$ +$CL_{intramembranous}$ = 0.00893 L/h/kg). This flow was assumed as a system parameter and applied across the different compounds.

Before predicting the drug kinetics during pregnancy, all compound files were first built and verified for their performance in nonpregnant populations [22]. Once the PBPK model adequately predicted the kinetics in nonpregnant subjects, the settings were retained to predict drug kinetics in pregnant women at different gestational weeks. Predictions in all cases were performed where the virtual population demographics and simulation settings/designs matched the original studies. Default demographics and age ranges were used when those data were not provided in the original paper.

2.2.1 Acceptance Criteria

Predicted pharmacokinetic profiles and parameters were compared with the sets of clinical observations available in the literature. The PBPK model predictions were considered successful and accepted if the observed pharmacokinetic profile fell within the 95th and 5th percentiles of predicted data and the predicted pharmacokinetic parameters fell within the range twofold of the observed data.

2.3 Acyclovir

Acyclovir is an antiviral drug that is commonly used during pregnancy for the suppression and treatment of a herpes outbreak. Different clinical studies have evaluated the pharmacokinetics of acyclovir during late pregnancy and have reported the concentrations in maternal circulation [23–26], the umbilical cord, and amniotic fluid [23–26].

2.3.1 Model Building

A previously published PBPK model for acyclovir elimination kinetics using the MechKiM was used, where two renal transporters were used to describe the acyclovir elimination kinetics, namely the basal OCT2 uptake and the apical multidrug and toxic compound extrusion (MATE) efflux transporters [27]. Tissue Kps predicted using the Rodgers and Rowland method [17] were scaled up with a global Kp scalar of 1.7 (optimized using 2.5 mg/kg data after acyclovir infusion for 1 h [28]). Acyclovir absorption was then described using the first-order absorption rate constant (optimized using 400 mg acyclovir oral data [29]). To extend the nonpregnant model to predict acyclovir kinetics during pregnancy, the pregnancy population was used with the permeability-limited placenta model and the full PBPK fetal model being selected. While no informative data could be found to describe the changes in renal MATE transporters during pregnancy, data were available for changes in the activity of the renal OCT2 transporter [30–32]. The increase in OCT2 during pregnancy was incorporated into the model using Eq. (2):

$$\text{Renal OCT2 (fold change from nonpregnant)} = 1 \times \left(1 + 0.0406 \times \text{GW} - 0.0007 \times \text{GW}^2\right), \quad (2)$$

where GW is the maternal gestational week. To parameterize the full feto–placental model, data from an ex vivo human placenta perfusion experiment were used to describe the transplacental passage of acyclovir [33]. Acyclovir crosses the placenta by simple diffusion [33, 34]. There was no placental metabolism of the drug [33]. The quantified maternal-to-fetal clearance was found to be similar to the fetal-to-maternal clearance, with a mean value of 1.74 mL/h/g of placental tissue [33]. This clearance value was used directly in the model, assuming a unity density for the placenta, i.e., 0.00174 L/h/mL of placenta, and applied to clearances on both the maternal-faced (CL_{PDM}) and fetal-faced (CL_{PDF}) sides of the placenta. Fetal CL_R was calculated using the acyclovir CL_R value in adults and scaled per kilogram of fetal bodyweight (CL_R = 0.050 L/h/kg). Since the $CL_{intramembranous}$ is assumed to be bidirectional, the fetal CL_R value used in the model was a composite parameter ($CL_R + CL_{intramembranous}$ = 0.050 + 0.00417 = 0.054 L/h/kg fetal weight). Similarly, combined fetal $CL_{swallowing}$ and $CL_{intramembranous}$ (0.00893 L/h/kg fetal weight) was used in the model, as mentioned in Sect. 2.2.

Fetal Kps were predicted within the Simulator using the Rodgers and Rowland method [17] without any adjustment to the calculated tissue Kps.

The performance of the developed model was verified against different clinical studies. The full list of the virtual trial designs used for nonpregnant model building and in model predictions in nonpregnant and pregnant subjects after acyclovir administration is provided in Sect. 2.1 of the electronic supplementary material (ESM).

2.4 Emtricitabine

Emtricitabine is an antiretroviral drug used to treat HIV infection and prevent HIV transmission during pregnancy. It is primarily excreted unchanged in the urine, and its pharmacokinetics are similar in healthy volunteers and HIV-infected individuals [38]. Emtricitabine also readily crosses the blood–placenta barrier [39].

2.4.1 Model Building

A previously published emtricitabine PBPK compound file for a nonpregnant population [40] was used. This nonpregnant model was extended to include the full PBPK feto–placental model to predict drug exposure in the fetus. Data from an ex vivo placental perfusion experiment (De Sousa Mendes et al. [41]) were integrated within the fetal–maternal PBPK model to parameterize the emtricitabine transplacental passage. The reported emtricitabine transplacental transfer was 0.104 L/h for the perfused cotyledon [29], corresponding to a total clearance ($CL_{placenta}$) of 1.2 L/h for the whole placenta or 0.0018 L/h/mL of placenta volume using the reported cotyledon volume of 58 mL. The $CL_{placenta}$ per mL placental volume was used in the model to describe emtricitabine permeability on both sides of the placenta, i.e., CL_{PDM} and CL_{PDF}.

Fetal CL_R was calculated as described for acyclovir but using adult emtricitabine CL_R and combined with $CL_{intramembranous}$ (0.0369 + 0.00417 = 0.041 L/h/kg fetal weight). Similarly, the combined fetal $CL_{swallowing}$ and $CL_{intramembranous}$ (0.00893 L/h/kg fetal weight) was incorporated into the model as described in Sect. 2.2.

Predicted exposure in the nonpregnant and pregnant populations were compared with the observed data. All virtual trial designs used for predicting emtricitabine in nonpregnant and pregnant subjects after oral administration are provided in Sect. 2.2 in the ESM.

2.5 Lamivudine

Lamivudine is an antiviral drug used to treat patients with HIV, including HIV-infected pregnant women. Lamivudine is not significantly metabolized by cytochrome P450 enzymes, and only 5–10% of the oral dose is metabolized to an inactive sulfoxide metabolite. Its CL_R represents a mean of approximately 70% ± standard deviation 16% of the total clearance in subjects with normal renal function (see references in Johnson et al. [51]). Prospective reports from the Antiretroviral Pregnancy Registry recorded over 11,000 exposures to lamivudine during pregnancy resulting in live births (including over 4300 exposed in the first trimester). The rate of lamivudine-exposed overall birth defects did not differ from the background birth defect rate [52].

2.5.1 Model Building

The lamivudine compound file was developed from drug-specific parameters available in the literature. These input parameters are provided in Table 1 in the ESM. Clinical studies of the pharmacokinetics of lamivudine in pregnancy have shown that lamivudine diffuses freely across the placenta from maternal to fetal circulation [53–57]. Several studies have reported lamivudine concentrations in the maternal circulation, umbilical cord, and amniotic fluid, but only two studies reported concentration profiles in maternal plasma [53, 56] and one study reported the concentration profile in the umbilical cord and amniotic fluid [53]. The ontogeny function for the renal OCT2, described in Sect. 2.3.1, was incorporated in the lamivudine model. Data from an ex vivo placental perfusion experiment showed that

the lamivudine clearance index ($CLI_{lamivudine}$) was 0.23 and 0.14 at a maternal concentration of 1.39 and 14.68 µg/mL, respectively [58]. The clearance of lamivudine per cotyledon was calculated from a mean clearance index value of 0.185 using the reference antipyrine clearance per cotyledon value of 1.58 mL/min/cotyledon [34] according to Eq. (3) [59].

$$\text{ex vivo } CL_{lamivudine} = CLI_{lamivudine} \times \text{ex vivo } CL_{antipyrine} \quad (3)$$

This resulted in a $CL_{lamivudine}$ value of 0.022 mL/min/cotyledon. This value was then scaled per milliliter of placenta as previously explained for emtricitabine: $CL_{lamivudine}$ = 0.0004 L/h/mL of placenta volume. This value was then used to describe lamivudine clearance on CL_{PDM} and CL_{PDF}.

Fetal CL_R was calculated using the lamivudine CL_R value in adults and scaled as described for the acyclovir case study. The input value in the model was a composite parameter (lamivudine $CL_R + CL_{intramembranous}$ = 0.04767 + 0.00417 = 0.052 L/h/kg fetal weight). Similarly, the combined fetal $CL_{swallowing}$ and $CL_{intramembranous}$ (0.00893 L/h/kg fetal weight) was incorporated into the model as described in Sect. 2.3.1.

Predicted exposures in nonpregnant and pregnant populations were compared with the observed data. All virtual trials, including four virtual trial designs for the nonpregnant PBPK model building model prediction after lamivudine administration are given in Sect. 2.3 in the ESM.

2.6 Metformin

Metformin is an antidiabetic drug with potential for use in the management of gestational diabetes and for reducing the otherwise high rate of spontaneous abortion in women with polycystic ovary syndrome. Metformin drug disposition during pregnancy has been evaluated in a few studies, showing that the enhanced CL_R during pregnancy has a significant effect on maternal exposure [30, 31, 64–66]. Metformin reaches the umbilical cord at delivery in almost the same magnitude as in the maternal circulation, with high interindividual variability [30, 64].

2.6.1 Model Building

The default metformin compound file within the Simulator compound library was used. The compound file uses the metformin free base dose rather than the metformin salt, so we entered the reported doses as free base in the model (1 mg metformin-hydrochloride salt contains 0.78 mg metformin free base). The activity of renal OCT2 during pregnancy changes according to Eq. (2). $CL_{swallowing}$ = 0.00893 L/h/kg fetal weight and fetal CL_R = 0.0917 L/h/kg fetal weight were calculated as mentioned in Sect. 2.2 and combined with the $CL_{intramembranous}$, so the input value in the model was 0.0958 L/h/kg fetal weight. The permeability clearance on both CL_{PDF} and CL_{PDM} was parameterized using a value of 0.0006 L/h/mL placenta after correcting the unit to the reported value of 0.585 mg/h/g of cotyledon (six term placentas) obtained from ex vivo experiments [67].

Predicted exposures in nonpregnant and pregnant populations were compared with the observed data. All virtual trial designs used for model predictions after metformin administration are given in Sect. 2.4 of the ESM.

3 Results

Figure 1 shows the maternal–feto–placental model. The complexity of the maternal part of the whole model was varied from drug to drug, with acyclovir, lamivudine, and metformin using details from the MechKiM, replacing the single perfusion kidney compartment used for emtricitabine.

3.1 Acyclovir

Figure 2a shows the PBPK predictions for acyclovir in the plasma of nonpregnant subjects. Figure 2b provides the predictions of acyclovir exposure during pregnancy (plots A1–F1 for maternal systemic exposure, plots A2–F2 for umbilical vein). These results are in good agreement with the observed data. The observed data are within the predicted 5th–95th percentiles range. Table 1 provides a comparison of the predicted maternal acyclovir pharmacokinetic parameters obtained from the simulated trials and those available from clinical studies. Table 2 provides a comparison of the predicted acyclovir pharmacokinetic parameters obtained from the simulated trials with those available from clinical studies for the umbilical cord and amniotic fluid.

3.2 Emtricitabine

Figure 3a shows the PBPK predictions for emtricitabine in the plasma of nonpregnant subjects. Figure 3b shows predictions of emtricitabine exposure during pregnancy (plots A–E for maternal systemic exposure), with plots G1 and G2 showing emtricitabine during delivery for maternal plasma and umbilical cord plasma, respectively. These results show that the predicted profiles closely followed the observations in nonpregnant subjects and during pregnancy for maternal circulation and the umbilical cord. The predicted clearance at different gestational ages during pregnancy are in line with the observed data (Fig. 3f). Table 3 provides a comparison of the predicted maternal emtricitabine pharmacokinetic parameters obtained for the simulated trials with those available from clinical studies following single or multiple administration. Table 2 provides a comparison of the predicted emtricitabine pharmacokinetic parameters obtained

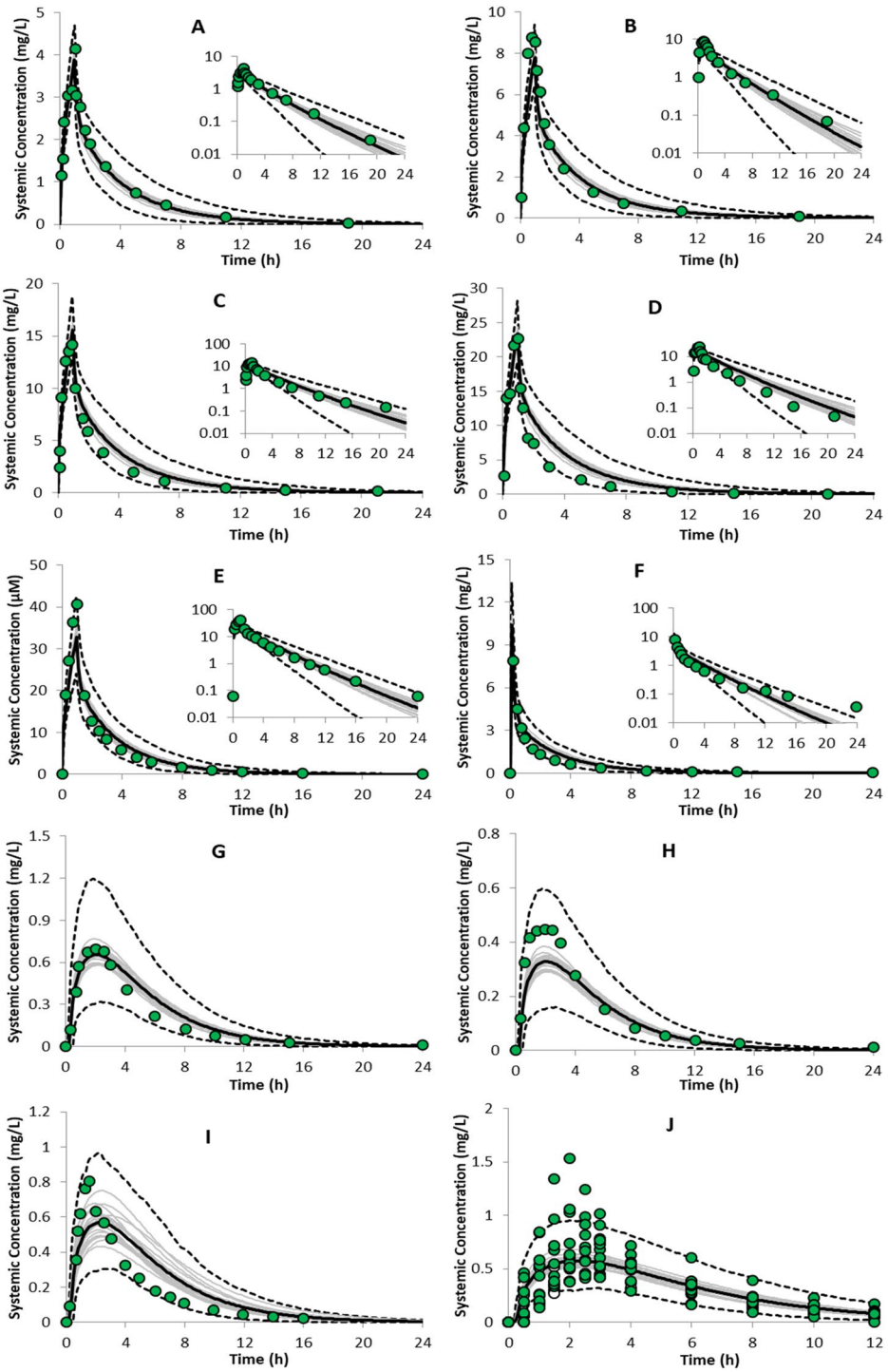


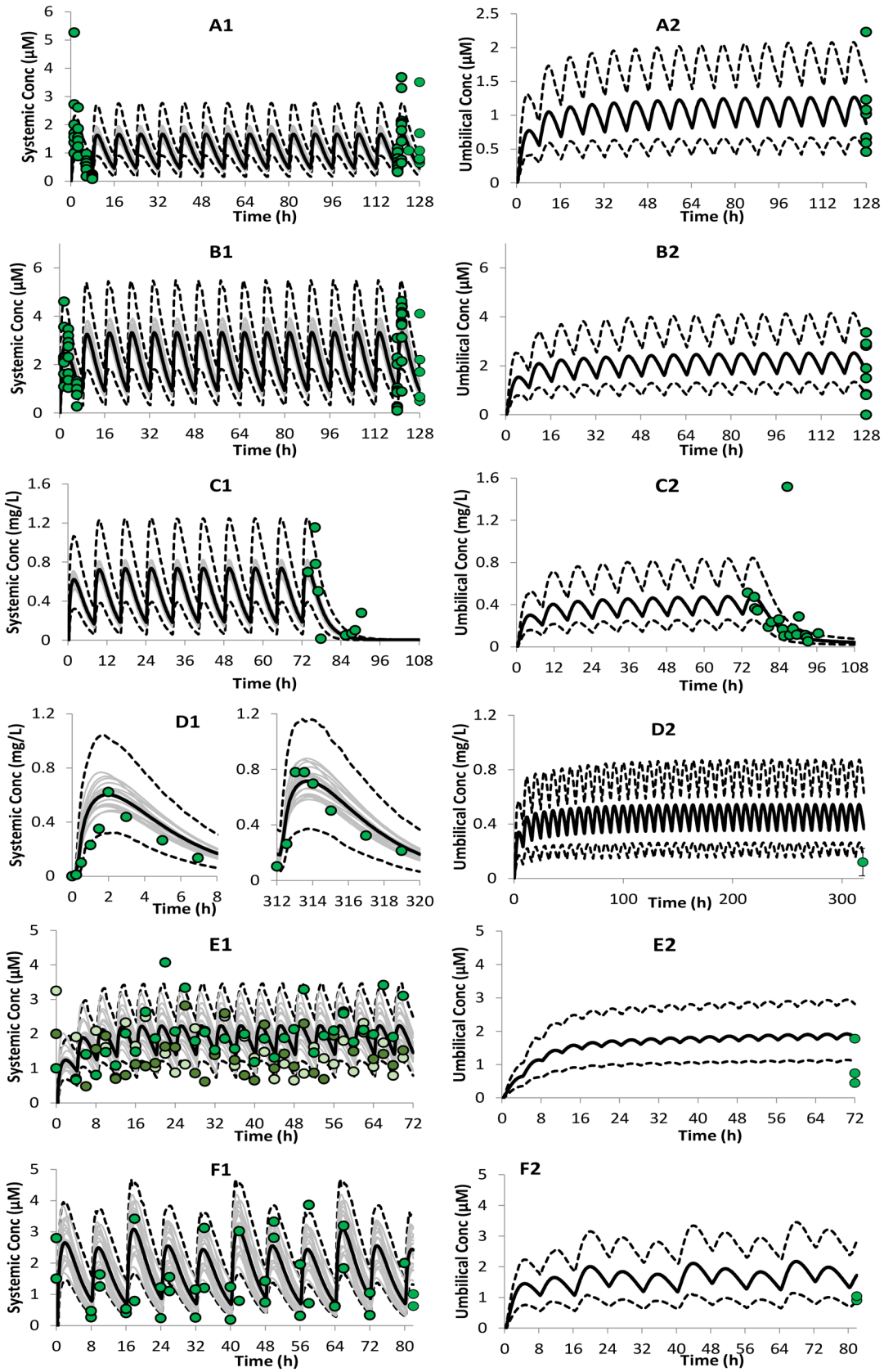
Fig. 2 Physiologically based pharmacokinetic model-predicted (lines) vs. observed (filled circles) acyclovir profiles in **A** nonpregnant and **B** pregnant subjects. **A** Profiles after a single intravenous dose of acyclovir 2.5 mg/kg (plot A), 5 mg/kg (plot B), 10 mg/kg (plot C), and 15 mg/kg (plot D) [28], and 350 mg (plot E) [35] infused over 1 h and 250 mg infused over 10 min (plot F) [29]. Profiles after oral administration of acyclovir 400 mg (plot G) and 200 mg (plot H) [29], 400 mg (plot I) [36], and 400 mg (plot J [circles are individual data]) [37]. **B** Profiles after multiple oral doses TID in maternal (200 mg TID [plot A1] [26], 400 mg TID [B1] [26], 400 mg TID [C1] [24], 400 mg TID [D1] [25], 200 mg/4 h [E1] [23], and variable dosing regimens [F1] [23]) and umbilical vein (A2 [26], B2 [26], C2 [24], D2 [25], E2 [23], F2 [23]). Bold continuous lines are the predicted means, thin gray lines are the trial means, and broken lines are the 5th and 95th percentiles. See Sect. 2 for trial settings. *Conc*. concentration, *TID* three times daily

from the simulated trials with those available from clinical studies in the umbilical cord and amniotic fluid.

3.3 Lamivudine

Figure 4a shows the predicted lamivudine concentration profiles for nonpregnant subjects after different routes of administration. Figure 4b provides model predictions for lamivudine during pregnancy and at labor (plots A, C, and G1 for maternal systemic exposure; plot D and G2 for umbilical vein; and plots B and E for amniotic fluid). These plots show good agreement between the PBPK predictions and the clinical observations. The predicted continuous rise in lamivudine clearance during pregnancy and decline toward term are in line with the observed data (Fig. 4b–f). Table 4 provides a comparison of observed and predicted maternal

Fig. 2 (continued)



pharmacokinetic parameters for lamivudine. Table 2 provides a comparison of the predicted lamivudine pharmacokinetic parameters obtained from the simulated trials with those available from clinical studies in the umbilical cord and amniotic fluid.

3.4 Metformin

Figure 5a and b provides the predicted metformin concentration profiles for nonpregnant women after oral administration. Model predictions for metformin at different gestational weeks during pregnancy and at labor are also given (C–G, Ia, and Ja for maternal systemic exposure, and plots H, Ib, and Jb for umbilical vein). These plots show good agreement between the PBPK predictions and the clinical observations. Table 5 provides a comparison of observed and predicted maternal pharmacokinetic parameters for metformin, and Table 2 provides a comparison of the predicted metformin pharmacokinetic parameters obtained from the simulated trials with those available from clinical studies in the umbilical cord.

4 Discussion

We used a population-based PBPK approach to assess the performance of a fetal–maternal PBPK model within the Simcyp Simulator to predict maternal and fetal exposure during pregnancy of four drugs that are primarily eliminated renally. The predictions were obtained without any fitting or adjustment to the physiological-related or compound-related parameters during pregnancy. The gestational-dependent physiological parameters and their interindividual variability [9–13] incorporated within the model facilitated the prediction of drug dispositions at different gestational weeks. The whole feto–placental model, and its parameters for the assessed compounds, were completely developed using a bottom-up approach with data from ex vivo human placental experiments together with physiological parameters. The ex vivo placental perfusion experiment parameters, when integrated within the mechanistic fetal–maternal PBPK model, provided a reliable source for parameterizing the transplacental passage of drugs and facilitated predictions of their fetal exposure.

During a normal healthy pregnancy, the maternal kidneys and their functions undergo time-dependent changes [9]. By term, kidney dimensions have increased by approximately 1 cm, and the volume has increased by as much as 30% [69, 70]. The average effective renal plasma flow increases from 32 L/h in nonpregnant women to 42 L/h at term, which corresponds to an effective renal blood flow of 53 L/h in nonpregnant women and 64 L/h in pregnant women at term. Likewise, the average GFR, measured by inulin clearance in healthy women, increases from 114 mL/min in nonpregnant women to 156 mL/min in pregnant women at term. This increase in GFR is also associated with an increase in creatinine clearance, from a pre-pregnancy value of 98.3 to 124 mL/min at term, which reflects the observed decrease in the serum creatinine level from 0.80 mL/dL in nonpregnant women to 0.66 mL/dL in pregnant women at term (see Abduljalil et al. [9] for more details on the pattern of these changes and interindividual variability at different GWs).

The PBPK model accurately predicted the acyclovir pharmacokinetic parameters in nonpregnant and pregnant subjects (Fig. 2 and Table 1). All predicted pharmacokinetic parameters were in good agreement with values reported in the corresponding clinical studies. The developed model reproduced the observed umbilical exposures at delivery (Fig. 2b and Table 2). The model predicted that the cord:maternal acyclovir ratio in plasma based on area under the plasma concentration-time curve (AUC) was 0.94 (range 0.78–1.03), which agrees with the observed ratio based on the drug plasma concentration at delivery of 0.80–1.29 [23, 25, 26]. The model indicated an accumulation of acyclovir in amniotic fluid with an amniotic fluid:maternal AUC ratio of 4.71 (range 4.66–4.80), but no accumulation is observed in the fetus. This agrees with clinical observations, where the amniotic fluid:maternal concentration ratio ranges from 2.8 to 6.5 and the amniotic fluid concentration was found to be three to six times higher than the corresponding cord plasma levels [25, 26]. This was also supported by observations that gastric aspirates from newborns ($n = 3$) had acyclovir levels similar to those of the amniotic fluid [26]. The model predicted an average placental concentration of 0.92 μM (5th–95th percentiles: 0.5–1.65), which is in the range of four measured values (1.08, 1.31, 1.34, and 6.19 μM [26]) after acyclovir 200 mg three times daily at term.

The emtricitabine PBPK predictions showed good agreement with the clinical observations in nonpregnant and pregnant subjects (Fig. 3 and Table 3). As shown in Table 3 (and Fig. 3b, plot G), the predicted oral clearance (CL/F) during the third trimester was about 1.3-fold higher in pregnant women than in nonpregnant women. This result was in line with the observed 1.2-fold increase in CL/F in pregnant women compared with nonpregnant women [47, 48, 50] and about 1.3-fold higher than the observed CL/F in healthy volunteers [43]. Predicted umbilical emtricitabine concentrations at delivery were in agreement with the observed data [39] (Fig. 3b and Table 2). The observed median umbilical AUC_{24h} at delivery is 8.2 mg/L [39], whereas the predicted median AUC_{24h} in the umbilical cord is 13 mg/L (range 6–24 mg/L). The developed model predicted a mean cord:maternal plasma ratio at term of 1.0 for AUC_{24h}. This predicted ratio agreed with the observed ratios of 0.8 (range 0.1–1.44) [39], 1.2 (90% confidence interval [CI] 1.00–1.5) [47], and 1.63 (range 0.46–1.82) [48]. In vitro

Table 1 Predicted vs. observed pharmacokinetic parameters of acyclovir in nonpregnant subjects and during pregnancy

Study, population	Dose	GWs	C_{max} (mg/L)			t_{max} (h)			AUC (mg · h/L)			CL (L/h)		
			Obs	Pred	Ratio	Obs	Pred	Ratio	Obs	Pred	Ratio	Obs	Pred	Ratio
Nonpregnant														
Laskin et al. [28]	2.5 mg/kg IV, 1 h infusion	NA	NA	4.0 ± 0.5	NA	NA	1.0 ± 0.0	NA	12.6 ± 2.1	11.3 ± 3.5	0.9	16.5 ± 1.8	18.1 ± 6.4	1.1
	5 mg/kg IV, 1 h infusion	NA	NA	8.0 ± 1.0	NA	NA	1.0 ± 0.0	NA	23.2 ± 8.8	22.6 ± 7.0	1.0	17.2 ± 6.0	18.1 ± 6.4	1.1
	10 mg/kg IV, 1 h infusion	NA	NA	16 ± 2.0	NA	NA	1.0 ± 0.0	NA	36.2 ± 8.5	45.1 ± 14.0	1.2	18.5 ± 6.3	18.1 ± 6.4	1.0
	15 mg/kg IV, 1 h infusion	NA	NA	24 ± 3.0	NA	NA	1.0 ± 0.0	NA	43.2 ± 11.7	67.7 ± 21.0	1.6	23.8 ± 5.9	18.1 ± 6.4	0.8
Soul-Lawton et al. [35], HV	350 mg IV, 1 h infusion	NA	9.2 ± 1.9	7.5 ± 1.4	0.8	0.98 ± 0.07	1.00 ± 0.00	1.0	18.9 ± 3.2	19.6 ± 5.1	1.0	19.3 ± 4.1	19.2 ± 5.3	1.0
Vergin et al. [29], HV	250 mg IV, 0.16 h infusion	NA	9.4 ± 1.7	11.7 ± 1.8	1.2	0.19 ± 0.04	0.15 ± 0.0	0.8	10.5 ± 2.2	13.6 ± 3.5	1.3	25.0 ± 6.9	19.7 ± 5.7	0.8
	200 mg PO QD	NA	0.53 ± 0.21	0.34 ± 0.14	0.6	1.6 ± 0.7	2.2 ± 0.5	1.4	2.6 ± 0.9	2.2 ± 0.9	0.8	76.9	108 ± 45	1.4
	400 mg PO QD	NA	0.80 ± 0.26	0.67 ± 0.29	0.8	1.8 ± 0.6	2.2 ± 0.5	1.2	3.9 ± 1.3	4.3 ± 1.7	1.1	102.6	108 ± 45	1.1
Lewis et al. [36], HV	400 mg PO QD	NA	0.86 ± 0.34	0.58 ± 0.21	0.7	1.4 ± 0.3	2.4 ± 0.5	1.7	3.3 ± 1.1	4.1 ± 1.7	1.2	121.2	112 ± 50	0.9
Amini et al. [37], HV	400 mg PO QD	NA	0.78 ± 0.31	0.59 ± 0.21	0.8	2.6 ± 1.2	2.3 ± 0.5	0.9	3.9 ± 1.1	3.8 ± 1.4	1.0	102.6	119 ± 45	1.2
Pregnant														
Kimberlin et al. [25]	400 mg PO, first dose	36	0.7 ± 0.6	0.62 ± 0.23	0.9	2.4 ± 1.2	2.1 ± 0.43	0.9	2.6 ± 0.8	3.7 ± 1.4	1.4	NA	122 ± 47	NA
	400 mg PO TID (ss)	38	0.9 ± 0.7	0.72 ± 0.3	0.8	1.5 ± 0.5	1.8 ± 0.3	1.2	3.7 ± 1.5	3.7 ± 1.3	1.0	NA	120 ± 40	NA
Frenkel et al. [26], HV	200 mg PO TID (ss)	38	0.43 ± 0.23	0.34 ± 0.12	0.9	1.5–3.0	1.0–2.4	NA	NA	1.7 ± 0.63	NA	NA	133 ± 55	NA
	400 mg PO TID (ss)	38	0.74 ± 0.23	0.70 ± 0.24	0.9	1.5–3.1	1.0–2.4	NA	NA	3.44 ± 1.23	NA	NA	133 ± 54	NA

Data are presented as mean ± standard deviation or range unless otherwise indicated

AUC area under the plasma concentration–time curve, *CL* total clearance, C_{max} maximum plasma concentration, *GW* maternal gestational week, *HV* healthy volunteers, *IV* intravenous, *NA* not applicable, *Obs* observed, *PO* oral administration, *Pred* predicted, *QD* once daily, *ss* steady state, *TID* three times daily, t_{max} time to C_{max}

Table 2 Predicted vs. observed maternal and fetal parameter ratios of acyclovir, emtricitabine, lamivudine, and metformin in the umbilical cord and amniotic fluid

Study	Dose	GW	Cord/maternal			Amniotic fluid/maternal		
			Obs	Pred[a]	Ratio	Obs	Pred[a]	Ratio
Acyclovir								
Kimberlin et al. [25]	400 mg PO, TID (ss)	36	0.80	1.02 (0.96–1.09)	1.3	6.50	4.80	0.7
Frenkel et al. [26], HV	200 mg PO, TID (ss)	38	0.81 (0.61–0.99)	1.03 (0.97–1.09)	1.3	4.29	4.66	1.1
	400 mg PO, TID (ss)	38	0.92 (0.71–1.21)	0.89 (0.34–3.42)	1.0	2.78	4.66	1.7
Haddad et al. [23], HV	200 mg PO, q4h (ss)	37	0.96 (0.67–1.13)	0.99 (0.92–1.06)	1.0	NA	NA	NA
	300, 400, 300 mg PO at 12 a.m., 8 a.m., and 4 p.m.	37	1.29 (0.9–1.68)	0.78 (0.47–0.99)	0.6	NA	NA	NA
Emtricitabine								
Hirt et al. [39], HIV	400 mg single dose[b]	39 (33–40)	0.76 (0.09–1.44)	0.89 (0.79–0.96)	1.2	NA	NA	NA
Stek et al. [47], HIV	200 mg QD[c] (ss)	35 (31–38)	1.2 (1.0–1.5)	1.1 (1.0–1.12)	0.92	NA	5.3	NA
Colbers et al. [48], HIV	200 mg QD[b] (ss)	33 (28–38)	1.63 (0.46–1.82)	1.2 (1.0–1.3)	0.74	NA	5.4	NA
Lamivudine								
Benaboud et al. [53], HIV	300 mg PO QD[b]	36–40	1.22 (0.32–21.8)	1.28 (0.91–1.54)	1.0	6.23	7.0	1.1
Moodley et al. [56], HIV	300 mg PO BID	38	1.00	0.92 (0.80–1.10)	0.92	NA	4.5	NA
Yeh et al. [57], HIV	150 mg PO BID[d]	37–40	1.0 (0.91–1.18)	1.16 (1.10–1.25)	1.16	14.2	6.0	0.42
Mandelbrot et al. [55], HIV	150 mg PO BID[d]	39	1.06	1.26 (1.19–1.32)	1.2	4.82	6.8	1.4
Metformin-HCl								
Liao et al. [30], diabetics	500 mg or 1000 mg PO BID	38	1.4 ± 0.5	1.3 ± 0.1	0.93	NA	4.6	NA
de Oliveira Baraldi et al. [68], PCOS	850 mg PO BID[b]	36–40	0.7 (0.4–1.3)	0.7 (0.6–0.8)	1.0	NA	5.0	NA
Tertti et al. [66], PWD	1000 mg BID	36	0.96 (0.14–4.36)	0.93 (0.76–1.1)	1.0	NA	7.6	NA

Data are presented as mean (range) or mean ± standard deviation unless otherwise indicated

BID twice daily, *GW* maternal gestational week, *HCl* hydrochloride, *HIV* HIV-infected patients, *HV* healthy volunteers, *NA* not applicable, *Obs* observed, *PCOS* polycystic ovary syndrome, *PO* oral administration, *Pred* predicted, *PWD* patients with diabetes, *q4h* every 4 h, *QD* once daily, *ss* steady state, *TID* three times daily

[a]Predicted cord/maternal and predicted amniotic fluid/maternal ratios were calculated based on the area under the plasma concentration–time curve at ss for the predicted concentration profiles

[b]Median (range)

[c]Geometric mean (90% confidence interval)

[d]Median (interquartile range)

studies reported that emtricitabine is a substrate for renal MATE1 [71] and multidrug resistance protein (MRP)-1 [72] transporters. These transporters were not part of the non-pregnant model, and their changes during pregnancy are neither documented nor accounted for in the developed model. From Fig. 3B, plot F, the predicted change in emtricitabine CL/F throughout pregnancy was adequate. Comparing the predicted and observed AUC and CL/F values during pregnancy (Table 3) also showed good prediction, i.e., within 1.25-fold. This was further supported by a sufficient prediction of the observed concentration profiles. While the integrated ex vivo placental data gave a good prediction of the umbilical exposure, the developed model did not take any transporter-mediated kinetics in the placenta into account. If such data become available from in vitro cell lines or other experimental settings, they can be incorporated in the model.

The developed lamivudine feto–maternal PBPK model adequately predicted the observed maternal and fetal pharmacokinetic profiles and parameters reported in the literature (Fig. 4 and Tables 2 and 4). In one study, the reported profiles for maternal, umbilical, and amniotic fluid concentrations included observations from lamivudine 150 mg twice daily and 300 mg once daily at steady state [53], so simulations were also executed for these two dosage regimens and the results combined. The current lamivudine model provided an expansion of the previously reported maternal PBPK model [73] by adding the detailed feto–placental and the MechKiM models as well as information from an ex vivo placental study. The model predicted a mean cord plasma:maternal plasma AUC ratio at birth of 1.2, which is in agreement with the observed ratios of 1.0 [56, 57], 1.22 [53], and 1.1 [55]. Lamivudine transplacental transporter

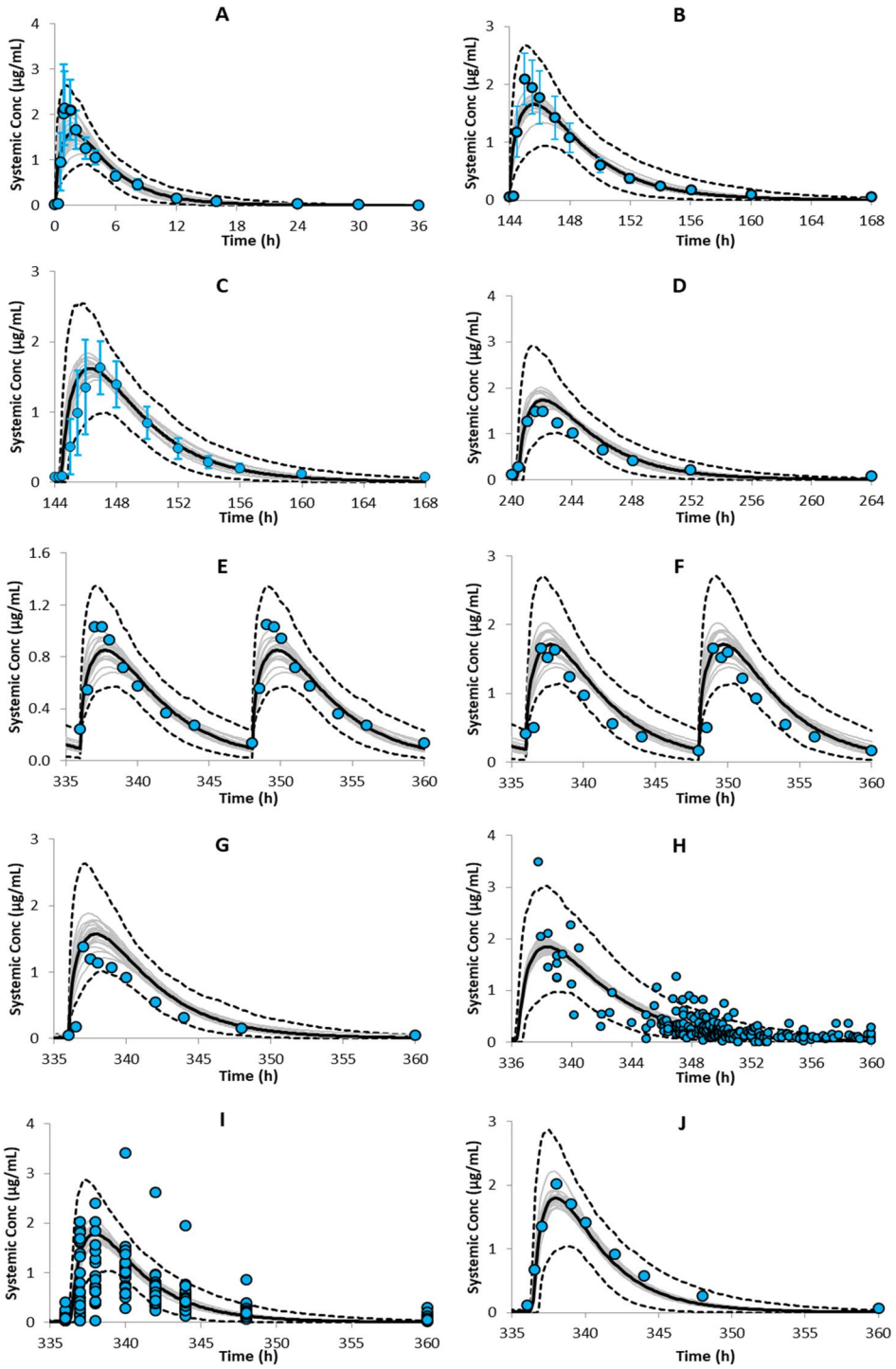


Fig. 3 Physiologically based pharmacokinetic model-predicted (lines) vs. observed (filled circles/squares) emtricitabine profiles in **a** nonpregnant and **b** pregnant subjects. Profiles after oral doses of emtricitabine: single (200 mg [plot A] [42]) or multiple (200 mg QD [plot B] [42]), 200 mg QD [plot C] [43], 200 mg QD [plot D] [44], 100 mg BID [plot E] [45], 200 mg BID [plot F] [45], 200 mg QD [plot G] [45], 200 mg QD [plot H] [46], 200 mg QD [plot I] [47], and 200 mg QD [plot J] [48]). **b** Profiles after multiple oral doses of emtricitabine 200 mg QD (A [48], B [47], C–E [49]). Plot F is oral clearance at different gestational weeks [50]. Plots G1 and G2 are for a single dose of emtricitabine 400 mg during labor [39]. Bold continuous lines are the predicted means, thin gray lines are the trial means, and broken lines are the 5th and 95th percentiles. See Sect. 2 for trial settings. *BID* twice daily, *Conc*. concentration, *QD* once daily

kinetics are still lacking. An in vitro study suggested that the fetal:maternal MATE1 efflux transporter might be involved in transplacental disposition of lamivudine [74]. In that study, the involvement of other placental transporters such as P-glycoprotein, breast cancer resistance protein, and MRP2 were suggested not to be relevant.

The metformin model predicted a 20% reduction in maternal exposure during pregnancy associated with a 20% increase in apparent total clearance due to a similar level of enhancement of the CL_R (Table 5). However, the predicted metformin half-life remained at the same level as in nonpregnant women (~3h), which is in agreement with the observed data [31]. The model predicted an umbilical vein:maternal plasma ratio of 1.1 (range 0.73–1.4) based on AUC at steady state, suggesting significant placental transfer of metformin. This prediction agrees with the reported ratio

Fig. 3 (continued)

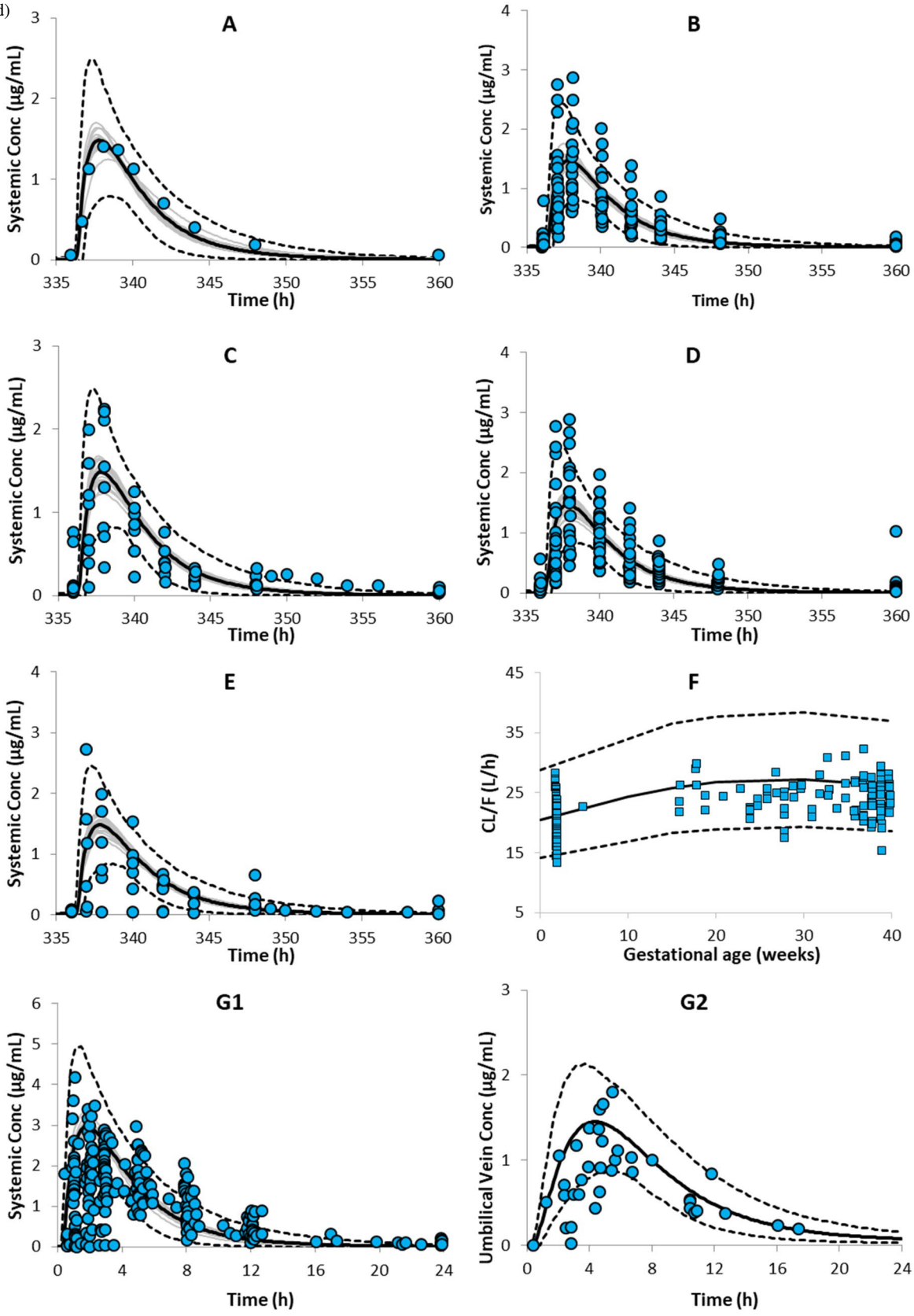


Table 3 Predicted vs. observed pharmacokinetic parameters of emtricitabine in nonpregnant subjects and during pregnancy

Study	Dose	GW	C_{max} (mg/L)			t_{max} (h)			AUC (mg · h/L)			CLpo (L/h)		
			Obs	Pred	Ratio	Obs	Pred	Ratio	Obs	Pred	Ratio	Obs	Pred	Ratio
Nonpregnant														
Zong et al. [42], HV	200 mg single dose	NA	2.36 (32%)[a]	1.67 (36%)[a]	0.7	1.21 (0.75–1.5)[b]	2 (0.55–5.9)[b]	1.7	11.3 (16%)[a]	10.5 (20%)[a]	0.9	NA	19.8 (21%)[a]	NA
	200 mg QD (ss)	NA	2.18 (21%)[a]	1.73 (33%)[a]	0.8	1.24 (0.98–2.0)[b]	1.85 (0.5–4.8)[b]	1.5	10.7 (16%)[a]	10.1 (21%)[a]	0.9	NA	20.7 (21%)[a]	NA
Blum et al. [43], HV	200 mg QD (ss)	NA	1.77 (22%)[a]	1.69 (31%)[a]	1.0	3.02 (29%)[a]	2.56 (29%)[a]	0.8	10.2 (19%)[a]	10.5 (20%)[a]	1.0	20.4 (23%)[a]	19.8 (21%)[a]	1.0
Wang et al. [44], HIV	200 mg QD (ss)	NA	1.85 (36%)[a]	1.82 (36%)[a]	1.0	1.88 (52%)[a]	2.4 (27%)[a]	1.3	10.0 (31 %)[a]	10.5 (20%)[a]	1.1	20 (31%)[a]	19.8 (21%)[a]	1.0
Valade et al. [50], HIV	200 mg QD (ss)	NA	NA	1.8 (0.5–5)[c]	NA	NA	2.2 (0.9–4.4)[c]	NA	9.8 (7.1–15)[c]	9.8 (5.7–17)[c]	1.0	20.5 (13.5–28)[c]	20.4 (12–35)[c]	1.0
Stek et al. [47], HIV	200 mg QD (ss)	NA	1.4 (1.3–1.7)[d]	1.8 (1.7–1.8)[d]	1.3	2 (1–6)[b]	2.3 (0.9–4.4)[b]	1.2	9.7 (8.6–10.9)[d]	9.8 (9.7–10)[d]	1.0	20.6 (18.4–23.2)[d]	20.4 (20–20.7)[d]	1.0
Colbers et al. [48], HIV	200 mg QD (ss)	NA	2 (1.8–2.3)[e]	1.8 (1.7–1.8)[e]	0.9	2 (1.0–6.0)[e]	2.2 (2.1–2.2)[e]	1.1	13 (11.8–14.3)[e]	9.9 (9.7–10)[e]	0.8	15 (14–17)[e]	20.3 (20–20.7)[e]	1.4
Pregnant														
Valade et al. [50], HIV	200 mg QD (ss)	15–28[c]	NA	1.5 (0.5–3.4)[c]	NA	NA	2.0 (0.7–3.8)[b]	NA	8.4 (6.7–10.9)[c]	7.5 (3.5–14.7)[c]	0.9	23.9[c]	26.6 (13.6–57)[c]	1.1
	200 mg QD (ss)	28–40[c]	NA	1.5 (0.4–3.7)[c]	NA	NA	2.0 (0.7–4.2)[b]	NA	8.2 (6.5–9.9)[c]	7.4 (3.4–14.4)[c]	0.9	24.5[c]	26 (15–56)[c]	1.1
	200 mg QD (ss)	At labor, 39[c]	NA	1.5 (0.5–3.5)[c]	NA	NA	2.0 (0.7–3.9)[b]	NA	8.3 (6.2–13.2)[c]	7.7 (3.6–15)[c]	0.9	24.1[c]	26 (13–56)[c]	1.1
Colbers et al. [48], HIV	200 mg QD (ss)	33 (28–38)[e]	1.8 (1.6–2.0)[e]	1.5 (1.4–1.5)[e]	0.8	2 (0.5–4.0)[e]	2 (1.96–2.1)[e]	1.0	9.6 (9.0–10.5)[e]	7.6 (7.4–7.7)[e]	0.8	21 (19–22)[e]	26.4 (26–27)[e]	1.3
Stek et al. [47], HIV	200 mg, QD (ss)	35 (31–38)[d]	1.4 (1.2–1.6)[d]	1.5 (1.4–1.5)[d]	1.1	2 (1–4)[b]	2.1 (0.7–4.4)[b]	1.1	8 (7.1–8.9)[d]	7.5 (7.4–7.6)[d]	0.9	25 (22.6–28.3)[d]	26.6 (26.2–27)[d]	1.1
Hirt et al. [39], HIV	400 mg, single dose	39 (33–42)[b]	1.68 (0.8–2.1)[b]	3 (0.6–7.4)[b]	1.8	NA	2.1 (0.7–4.6)[b]	NA	14.3 (11–19)[b]	15.2 (7.2–27.5)[b]	1.1	NA	26 (15–56)[b]	NA

AUC area under the plasma concentration–time curve, *CLpo* clearance after oral administration, C_{max} maximum plasma concentration, *GW* maternal gestational week, *HIV* HIV-infected patients, *HV* healthy volunteers, *NA* not applicable, *Obs* observed, *Pred* predicted, *QD* once daily, *ss* steady state, t_{max} time to C_{max}

[a]Mean (coefficient variation)

[b]Median (minimum–maximum)

[c]Geometric mean (minimum–maximum)

[d]Geometric mean (90% confidence interval)

[e]Geometric mean (95% confidence interval)

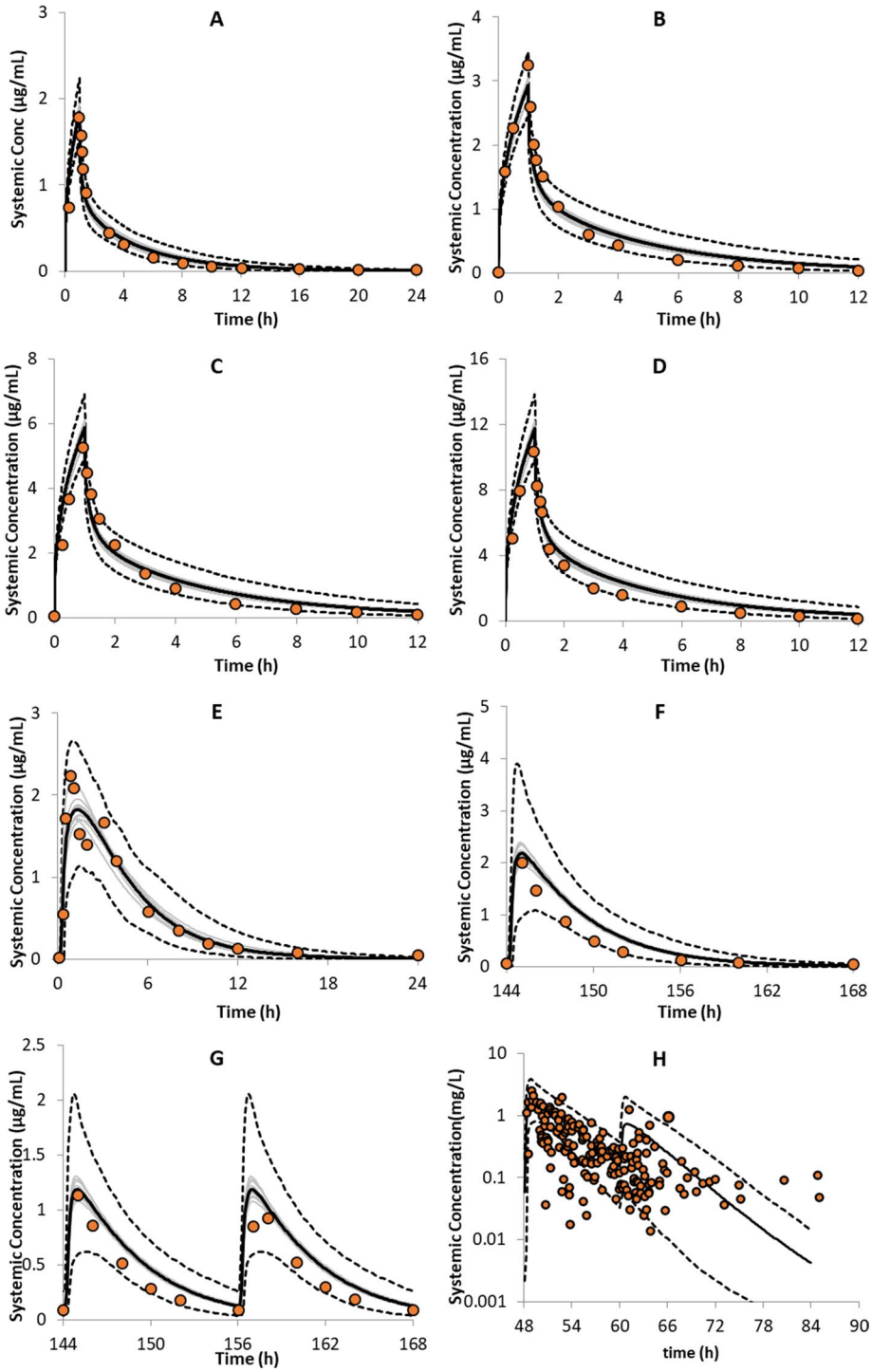


Fig. 4 Physiologically based pharmacokinetic model-predicted (lines) vs. observed (filled circles/squares) lamivudine profiles in **a** nonpregnant and **b** pregnant subjects. Profiles after a single intravenous dose of lamivudine (100 mg over 1 h [plot A] [60], 2 mg/kg over 1 h [plot B] [61], 4 mg/kg over 1 h [plot C] [61], and 8 mg/kg over 1 h [plot D] [61]), a single oral dose of lamivudine 300 mg (plot E) [62], and multiple oral doses of lamivudine 300 mg QD (plot F) [63] or 150 mg (plot G) [63], and mixed doses of lamivudine 150 mg BID and 300 mg QD (plot H) [53]. **b** Profiles after single (plot A) [56] and multiple (plot B) [56] oral doses of lamivudine 150 mg BID. Plots C, D, and E are after multiple oral doses of lamivudine 150 mg and 300 mg QD during labor (C, maternal plasma; D, umbilical vein; E, amniotic fluid) [53]. Oral clearances at different gestational weeks are given in plot F [53]. Multiple oral doses of lamivudine 150 mg BID at term [55]. Horizontal lines in plot G1 and G2 are the median and range of measured lamivudine concentrations. Bold continuous lines are the predicted means, thin gray lines are the trial means, broken lines are the 5th and 95th percentiles, and the error bars represent the standard deviations for the observed data. See Sect. 2 for trial settings. *BID* twice daily, *Conc*. concentration, *QD* once daily

taken opportunistically as paired concentrations measured at birth: 1.4 ± 0.5 [30], 0.96 (range 0.14–4.36) [66], and 0.7 (range 0.4–1.3) [68]. Predicted umbilical concentrations in the developed model overlapped with the observed data (Fig. 5, plots H, Ib and Jb). Umbilical exposure data for metformin are limited, with almost no data available on metformin concentrations in the amniotic fluid. The observed umbilical:plasma ratio of 0.96 (range 0.14–4.36) at term after maternal oral intake of metformin 1000 mg twice daily [66] agrees with the predicted ratio of 0.93 (range 0.76–1.1). This ratio was slightly lower after a dose of 850 mg twice daily, at around 0.7 (range 0.4–1.3) [68], which is also in agreement with the predicted ratio of 0.7 (range 0.6–0.8). Metformin is known a substrate for OCT1 and OCT2, which are responsible for the hepatic and renal transport of metformin but not for OCT3 [75], which is the only abundant

Fig. 4 (continued)

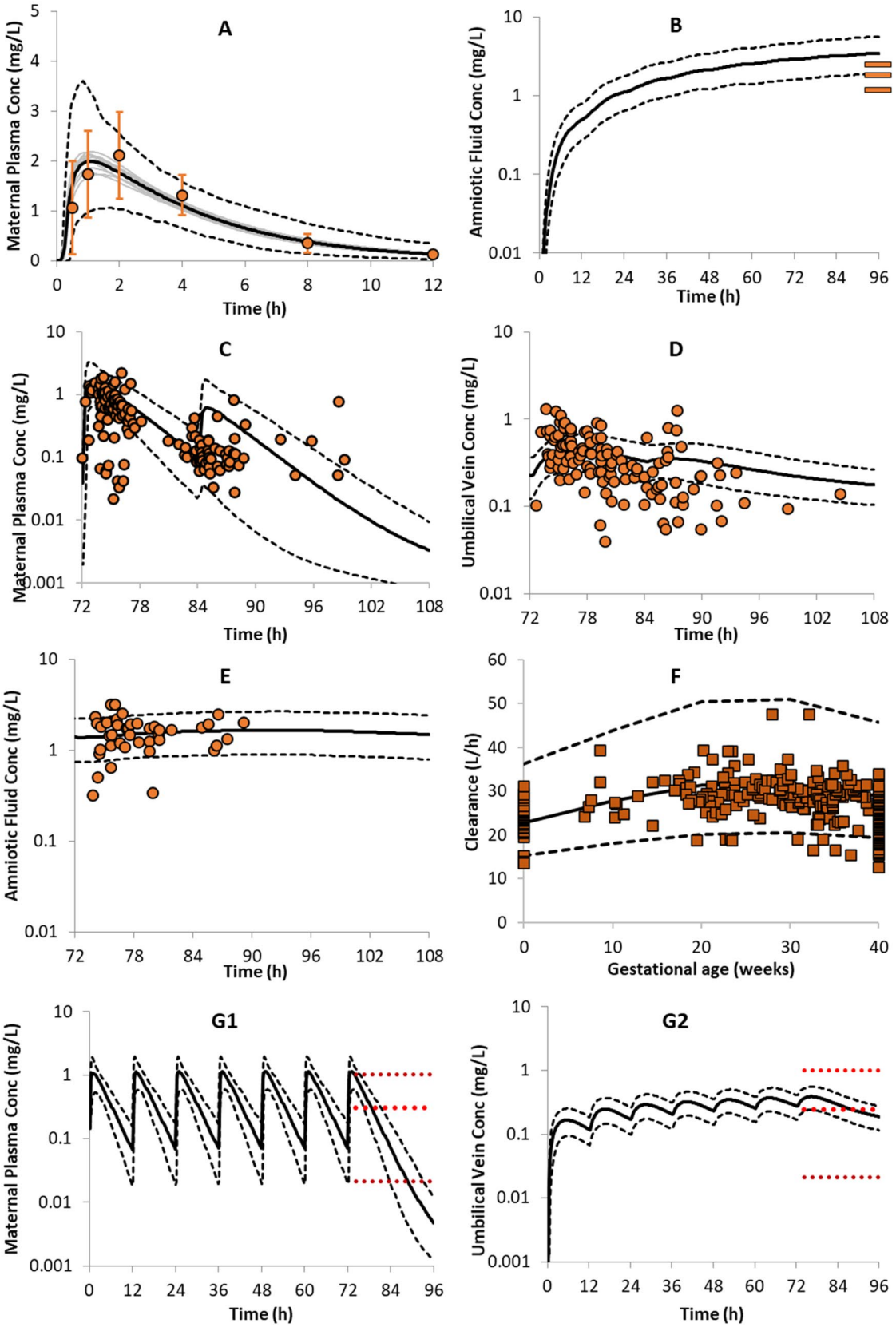


Table 4 Predicted vs. observed pharmacokinetic parameters of lamivudine in nonpregnant subjects and during pregnancy

Study	Dose (mg)	GW	C_{max} (mg/L)			t_{max} (h)			AUC (mg · h/L)			CL (L/h)		
			Obs	Pred	Ratio	Obs	Pred	Ratio	Obs	Pred	Ratio	Obs	Pred	Ratio
Nonpregnant														
Van Leeuwen et al. [61], HIV	2 mg/kg IV, 1 h infusion[a]	NA	3.3	2.9	0.9	0.98	1.00	1.0	6.7	7.1	1.1	21.7	22.9	1.1
	4 mg/kg IV, 1 h infusion[a]	NA	5.6	5.9	1.0	0.98	1.00	1.0	13.0	14.2	1.1	23.5	22.9	1.0
	8 mg/kg IV, 1 h infusion[a]	NA	10.6	11.8	1.1	0.98	1.00	1.0	22.8	28.5	1.2	25.9	22.9	0.9
Yuen et al. [63], HV	300 mg PO QD (ss)[b]	NA	2.0 ± 0.54	2.3 ± 0.9	1.2	1.13 ± 0.34	1.4 ± 0.7	1.2	8.9 ± 1.83	12.7 ± 3.3	1.4	33.7	25.5	0.8
	150 mg PO BID (ss)[b]	NA	1.2 ± 0.29	1.2 ± 0.5	1.0	1.19 ± 0.40	1.3 ± 0.5	1.1	9.3 ± 1.64	12.7 ± 3.3	1.4	32.3	25.5	0.8
Heald et al. [62], HIV	300 mg PO single dose[c]	NA	2.5	1.8	0.7	NA	NA	NA	10.9	9.6	0.9	27.5	31.3	1.1
Benaboud et al. [53], HIV	300 mg PO QD (ss)[d]	NA	NA	2.5 (0.84–8.3)	NA	NA	1.25(0.45–4.25)	NA	12.7	13.1 (4.8–25.6)	1.0	23.7	22.9 (11.7–61.9)	1.0
Pregnant														
Benaboud et al. [53], HIV	300 mg PO QD (ss)[d]	36–40	NA	2.2 (0.6–6.5)	NA	NA	1.2 (0.45–4.15)	NA	12.5	10.5 (3.8–22.3)	0.8	24.0	28.6 (13.5–79.1)	1.2
Moodley et al. [56], HIV	300 mg PO single dose	38[e]	2.5 (2.0–3.1)[e]	2.0 (1.8–2.1)[e]	0.8	1.5 (0.5–2.0)[d]	1.3 (0.5–5.3)[d]	0.9	10.0 (8.4–11.9)[e]	9.8 (9.4–10.2)[e]	1.0	32.5 (32.5–38.8)[e]	30.6 (29.5–31.8)[e]	0.9

AUC area under the concentration curve, *BID* twice daily, *CL* total clearance, C_{max} maximum plasma concentration, *GW* maternal gestational week, *HIV* HIV-infected patients, *HV* healthy volunteers, *IV* intravenous, *NA* not applicable, *Obs* observed, *PO* oral administration, *Pred* predicted, *QD* once daily, *ss* steady state, t_{max} time to C_{max}

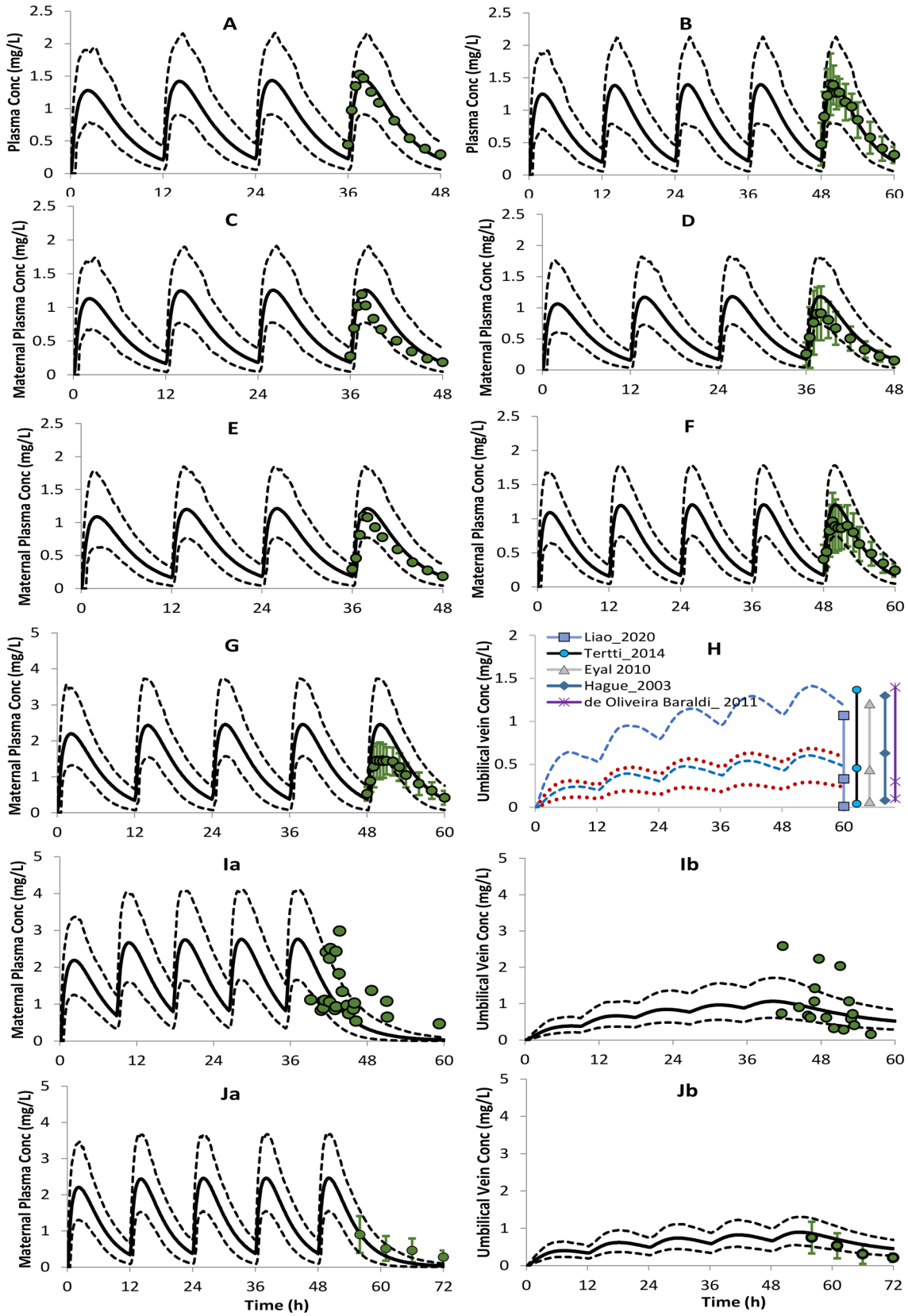
[a]Mean

[b]Mean ± standard deviation

[c]Geometric mean

[d]Median (range)

[e]Geometric mean (95% confidence interval)


A
B
C
D
E
F
G
H
Ia
Ib
Ja
Jb
Plasma Conc (mg/L)
Maternal Plasma Conc (mg/L)
Umbilical vein Conc (mg/L)
Umbilical Vein Conc (mg/L)
Time (h)
Liao_2020
Tertti_2014
Eyal 2010
Hague_2003
de Oliveira Baraldi_ 2011

◂**Fig. 5** Physiologically based pharmacokinetic model-predicted (lines) vs. observed (filled circles) metformin profiles in nonpregnant and pregnant women after multiple oral doses. Systemic plasma concentration profiles in nonpregnant women (plots A [31] and B [30]) and in maternal plasma at 12 GW (plot C) [31], 24 GW (plot D) [31], 36 GW (plot E) [31], 26–38 GW (plot F, 390 mg; plot G, 780 mg BID) [30] during delivery (plots Ia [64] and Ja [66]). Plot H is the predicted concentrations (5th and 95th percentiles for umbilical cord after multiple oral doses of 390 mg [dotted profile] and 780 mg [dashed profiles]) of metformin base. Vertical lines are observed umbilical concentrations at steady state, given as mean and ranges, obtained from Liao et al. [30] (squares), Tertti et al. [66] (circles), *a*nd Eyal et al. [31] (triangles), and as median and range (Hague et al. [88] [diamonds] and de Oliveira Baraldi et al. [68] (*crosses*). Plots Ib and Jb show the umbilical cord concentrations. Bold continuous lines are the predicted means, thin gray lines are the trial means, and broken lines are the 5th and 95th percentiles. Error bars, where applicable, represent standard deviations. See Sect. 2 for trial settings. *BID* twice daily, *Conc*. concentration, *GW* maternal gestational weeks, *QD* once daily

OCT isoform expressed in the human placenta [76, 77]. It is still unclear how metformin crosses the placental membrane. An ex vivo study showed that metformin does not transfer across the placenta solely by passive diffusion, and an active transport mechanism against the concentration gradient is involved, although not identified [78]. Another ex vivo study suggested that OCT transporters are not likely the transfer mechanism using cimetidine [79]. Cimetidine is a known inhibitor of OCT2, and MATE1 and OCTN2 transporters [80] showed no effect on metformin transfer from or to the fetal side of the placenta [79].

Researchers have published applications of PBPK to renally eliminated compounds during pregnancy [73, 81–84]. However, in most of these cases, the pregnancy PBPK models were built to replicate specific study designs and/or observations at a specific time point during pregnancy, and the fetus was represented by lumped compartments with limited physiological parameters [53]. Feto–placental PBPK models using ex vivo placental data have been published for emtricitabine [41, 49], acyclovir [49], acetaminophen [85], dolutegravir [86], darunavir [7], and nevirapine [6]. The current model expands the applicability of these lumped fetal models, adding different fetal organs.

The current work also represents the maternal kidney with a detailed mechanistic kidney model to allow inclusion of transporter kinetics and the gestational-dependent change in the activity of renal OCT2 transporters during pregnancy observed for acyclovir, metformin, and lamivudine. The contribution of this transporter to the overall CL_R varies during pregnancy (see Fig. S1 in the ESM). For emtricitabine, a simplified model using passive glomerular filtration was used. The lamivudine model adequately predicted the apparent total clearance during pregnancy and correctly predicted the clearance reduction trend towards the end of pregnancy, but with a slight overprediction (Fig. 4b, plot F; Table 4). This overprediction may suggest a possible increase in the activity of an efflux transporter on the apical kidney cells. However, data limitations meant this adjustment was not added into the model [56].

In the original studies, observed pharmacokinetic parameters were reported, depending on the type of analysis used, either the arithmetic mean, median, or geometric mean. Similarly, their variability was reported as ranges, standard error of the mean, standard deviation, interquartile range, 90th and 95th CIs, or coefficient of variability. These different methods of quantification further challenged the comparison of results between studies, so we reported the predicted pharmacokinetic parameters from the developed models in the observed unit for better comparison (Tables 1, 2, 3, 4). The original studies were also lacking information relating to the trial setting used and the inclusion criteria applied. This required us to replicate and compare individual studies separately. The absence of longitudinal studies, where individuals are their own control, further challenged our interpretation of the results.

Most of the original studies measured and reported the concentrations of these four drugs in the umbilical cord and amniotic fluid without reference to the exact sampling time, or reported the sampling time as a range spanning over 10 h after maternal drug administration; in a few cases, the data were lumped from different dosage regimens. Such data are of limited use for PBPK modeling to understand the mechanism of drug transfer and accumulation, if any. Another challenge was that many of the studies reported the cord:maternal plasma concentration ratio as a measurement of the extent of drug transferred via the placenta. Reporting the static concentration or cord:maternal plasma ratio using such crude metrics minimizes the information that can be extracted from such data. It could potentially provide a sensible index if only both maternal and fetal concentrations change in parallel with the same slope, otherwise this ratio would vary as the time elapsed between the last administration and the sampling time, which may warrant clinical concern. It may be more appropriate to report the raw data at delivery when the plasma can be collected at different time points after the last administered dose or the data can be analyzed to derive a reliable exposure parameter, such as AUC.

However, the developed PBPK model still has limitations. From a physiological perspective, the model does not consider the dynamic changes of the amniotic fluid volume. This is largely due to the currently poor understanding of the underlying mechanisms and how they are affected by the underlying disease state [87]. We used the ex vivo placental clearance as it was reported from the experiment on both sides of the placenta, without applying any in vitro–in vivo extrapolation techniques to these values, to account for experimental settings such as the binding components in the perfusates, differences in the flow rates, and other

Table 5 Predicted vs. observed pharmacokinetic parameters of metformin in nonpregnant and pregnant women

Study	Dose (mg)[a]	GW	C_{max} (mg/L)			t_{max} (h)			AUCss (mg · h/L)			CLpo (L/h)			Half-life (h)		
			Obs	Pred	Ratio	Obs	Pred	Ratio	Obs	Pred	Ratio	Obs	Pred	Ratio	Obs	Pred	Ratio
Nonpregnant																	
Eyal et al. [31] ($n = 6$)	500 mg PO BID	NA	1.61 ± 0.39	1.45 ± 0.43	0.9	1.8 ± 0.3	2.2 ± 0.4	1.2	9.8 ± 3.1	9.8 ± 2.9	1.0	55.7 ± 17.1	44.0 ± 16.1	0.8	3.8	3.0 ± 1.0	0.8
Liao et al. [30] ($n = 9$)	500 mg PO BID	NA	NA	1.41 ± 0.43	NA	2.1 ± 0.5	2.1 ± 0.4	1.0	9.8	9.4 ± 2.8	1.0	51.2 ± 13.8	45.6 ± 16.6	0.9	4.0 ± 0.8	3.0 ± 1.0	0.8
Liao et al. [30] ($n = 14$)	1000 mg PO BID	NA	NA	2.91 ± 0.89	NA	2.2 ± 1.3	2.2 ± 0.4	1.0	9.9	19.2 ± 5.9	1.9	100.7 ± 34.4	44.6 ± 15.5	0.4	3.7 ± 0.6	2.9 ± 0.9	0.8
Pregnant																	
Eyal et al. [31] ($n = 5$)	500 mg PO BID	10–14	1.22 ± 0.47	1.27 ± 0.39	1.0	1.4 ± 0.2	2.0 ± 0.4	1.4	6.5 ± 2.8	8.2 ± 2.5	1.3	87.4 ± 33.6	52.4 ± 20.6	0.6	4.0	3.0 ± 1.0	0.7
Eyal et al. [31] ($n = 7$)	500 mg PO BID	22–26	1.06 ± 0.49	1.20 ± 0.37	1.1	2.0 ± 0.7	2.0 ± 0.4	1.0	6.1 ± 3.4	7.7 ± 2.5	1.3	97.7 ± 38.9	55.9 ± 19.0	0.6	3.8	3.0 ± 1.0	0.8
Eyal et al. [31] ($n = 4$)	500 mg PO BID	34–38	1.14 ± 0.30	1.23 ± 0.37	1.1	2.0 ± 0.7	2.0 ± 0.4	1.0	6.9 ± 1.8	8.1 ± 2.4	1.2	77.0 ± 25.1	52.9 ± 17.0	0.7	3.9	3.1 ± 1.0	0.8
Liao et al. [30] ($n = 39$)	500 mg PO BID	26–38	NA	1.22 ± 0.34	NA	2.6 ± 1.3	2.0 ± 0.4	0.8	7.7	7.9 ± 2.3	1.0	65.1 ± 19.7	53.7 ± 15.6	0.8	4.2 ± 1.1	3.0 ± 1.0	0.7
Liao et al. [30] ($n = 15$)	1000 mg PO BID	26–38	NA	2.50 ± 0.71	NA	2.4 ± 1.1	2.1 ± 0.4	0.9	11.9	16.5 ± 5.0	1.4	84.3 ± 23.0	51.7 ± 15.6	0.6	4.3 ± 0.7	3.0 ± 0.9	0.7

Data are presented as range or mean ± standard deviation

AUCss area under the plasma concentration–time curve at steady state, *BID* twice daily, *CLpo* clearance after oral administration, C_{max} maximum plasma drug concentration , *GW* maternal gestational week, *n* sample size, *NA* not applicable, *Obs* observed, *PO* oral administration, *Pred* predicted, t_{max} time to C_{max}

[a]Metformin dose as hydrochloride salt

experimental variables. An approach to using the ex vivo clearance mechanistically is yet to be developed. The predicted variability in the current feto–placental model stems from the variability in the placenta and fetal bodyweight only, therefore the observed variability in fetal, placental, and amniotic fluid pharmacokinetic parameters were lower than the observed variability. This discrepancy can be assessed by adding random variability to the fetal CL_R, $CL_{swallowing}$, and CL_{PD} on both sides of the placenta in the model; however, assessing different levels of variability in the developed model was not considered as part of the scope of this work. Changes in the activity of placental transporters during pregnancy were not included in this model. The transplacental passage of the four drugs used data from an ex vivo experiment that measured the net flux of drug, i.e., included both the passive process and any involved efflux or uptake placental transporters. We did not include direct drug diffusion between the maternal circulation and amniotic fluid as the relevant data were missing.

To build confidence in the model, further verification of the feto–placental model for drugs undergoing metabolism with compounds of diverse properties will be required. Comparable work has been done using maternal PBPK models [22]. Similarly, the evaluation of compounds for which placental transporters play an important role in fetal exposure is required to understand the magnitude of changes in placental transporters during pregnancy. Expanding the scope of the pregnancy PBPK model applications to include additional enzymes and transporters is essential to obtain a better insight into the effects of gestation on drug disposition during pregnancy.

5 Conclusion

We developed a detailed maternal–fetal PBPK model that successfully predicted the drug kinetics during pregnancy of four renally eliminated compounds. The modeling process was informed using the diffusion clearance around the blood–placenta barrier from ex vivo cotyledon experiments without any refinement of model parameters. The predictions demonstrate the utility of pregnancy PBPK modeling as a generic tool for predicting the maternal and fetal kinetics of renally excreted drugs and can be used to assess dosage requirements. The fetal PBPK model should be viewed as a "live" model that will undergo continuous enhancement and improvement as knowledge of the mechanisms of drug distribution and elimination within the feto–placental unit increases. While the level of confidence in these models at any given time reflects the state of existing knowledge and the capability to incorporate such knowledge, the models can play an important role in understanding drug safety and efficacy for both the mother and the developing fetus. The proposed PBPK approach is promising and can inform the design of clinical pharmacokinetic studies and assist with predicting maternal and fetal drug exposures at different stages of pregnancy.

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Declarations

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Conflicts of interest Khaled Abduljalil, Amita Pansari, Jia Ning, and Masoud Jamei are full-time employees of Certara UK Limited, Simcyp Division. The activities of Certara are supported by a consortium of pharmaceutical companies. The Simcyp Simulator is available at no cost, following completion of the training workshop, to approved members of academic institutions and other not-for-profit organizations for research and teaching purposes.

Ethics approval Not applicable.

Consent to participate Not applicable.

Availability material Not applicable.

Consent for publication Not applicable.

Code availability Not applicable.

Author contributions KA was responsible for the conception and design of the study. KA, JN, and AP performed the experiments and acquired the data. KA, AP, JN, and MJ analyzed and interpreted the data. KA drafted or revised the manuscript critically for important intellectual content with input from all authors.

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